

# Centering, Mediators, and Residuals

Teaching note

Callen, Livnat, and Segal (2009), *The Accounting Review*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

## Three designs on one dataset

Callen, Livnat, and Segal ask whether quarterly earnings, and their cash flow and accrual components, are priced in single-name CDS premia (2002-2005; 12,242 contracts on 423 firms), where prior levels-only studies disagreed on the sign.

*“The CDS is essentially a ‘pure credit’ default instrument, providing a far less noisy measure of credit risk in comparison to other debt instruments.”*

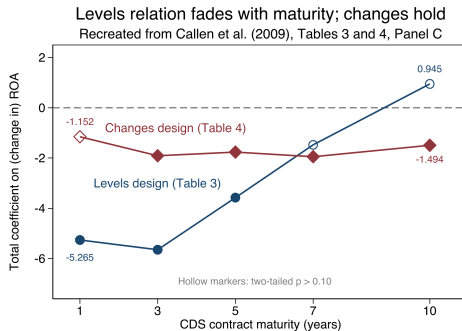
Callen, Livnat, and Segal (2009, p. 1364)

The levels design (log premium one day after the SEC filing, Table 3) and the changes design (filing-to-filing percentage change, Table 4) share the structural determinants (leverage, volatility, the risk-free rate, maturity), the S&P rating or its change, and restructuring clauses; the three-day announcement window (Table 5) keeps only the maturity and restructuring indicators. All three use firm random effects.

## The maturity pattern rests on interactions

H1 through H3 predict an inverse earnings-premium relation that weakens with maturity; each Panel C total is the base ROA coefficient plus an  $\text{ROA} \times \text{maturity}$  interaction.

In levels the relation is significant through five years and insignificant at seven and ten ( $p = 0.212$  and  $0.424$ ); in changes it is flat across maturities and insignificant only at one year.



## Centering is a reparameterization

*“To reduce potential multicollinearity induced by interaction terms, we de-mean all continuous independent variables (Aiken and West 1991).”*

Callen, Livnat, and Segal (2009, p. 1373)

Panel B model, continuous  $x$  (ROA) and maturity indicator  $D$ ; substitute  $x = x_c + \bar{x}$  with  $x_c$  demeaned:

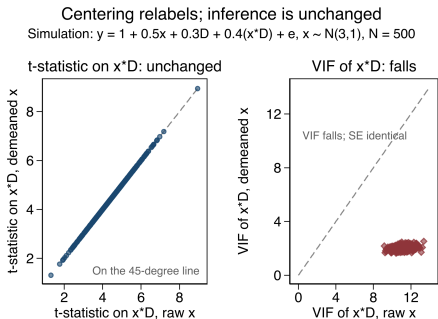
$$y = \beta_0 + \beta_1 x + \beta_2 D + \beta_3 xD + e = (\beta_0 + \beta_1 \bar{x}) + \beta_1 x_c + (\beta_2 + \beta_3 \bar{x}) D + \beta_3 x_c D + e$$

The two regressor sets span the same column space, so fitted values, residuals,  $R^2$ ,  $\beta_1$ ,  $\beta_3$ , and their standard errors are identical. Only the intercept and the coefficient on  $D$  are relabeled (the maturity effect is now evaluated at mean ROA). The VIF describes the columns; the standard error describes the estimate (Echambadi and Hess 2007; Kromrey and Foster-Johnson 1998).

## Centering leaves the $t$ -statistic unchanged

Across 500 samples the largest difference in the interaction  $t$ -statistic is  $5 \times 10^{-7}$ ; the mean VIF of the interaction regressor falls from 11.1 to 2.0 because its own variance falls by the same factor.

In Table 3 Panel B only ROA enters an interaction, so demeaning can alter only the intercept and the maturity dummies; the interpreted coefficients are unchanged.



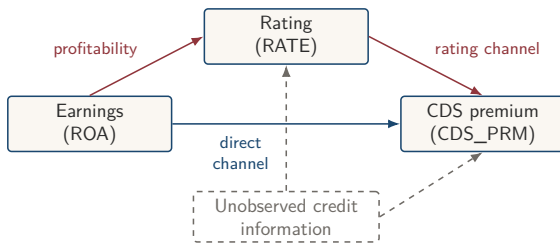
## The rating lies between earnings and premia

H1 (p. 1369) states that the premium “is inversely related to the reference entities’ quarterly earnings,” and the headline magnitudes (pp. 1364, 1392) are stated as effects of earnings: total effects. Yet RATE is a regressor in every levels column.

*“Credit ratings provide a noisy measure of credit risk and are employed in all the empirical CDS studies referenced above.”*

Callen, Livnat, and Segal (2009, p. 1373)

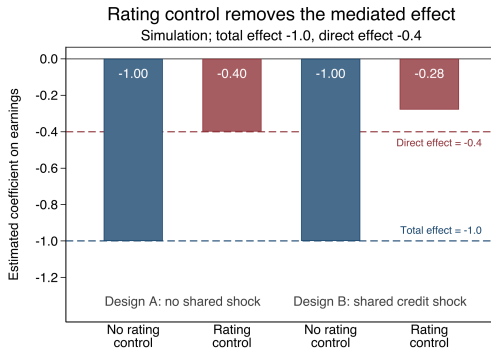
Conditioning on RATE closes the maroon path: the ROA coefficient becomes the direct effect, consistent only if nothing unobserved moves both rating and premium (Cinelli, Forney, and Pearl 2022).



## The mediator control changes the estimand

Without the rating, the regression recovers the total effect. With the rating, it recovers the direct effect only when no unobserved variable enters both the rating and the premium, and neither quantity otherwise.

The paper conditions on the mediator in Table 3 (RATE), Table 4 (rating changes), and Table 6 (the BBB-and-below partition); earnings beyond the rating is not the question H1 asks.



## Table 5 orthogonalizes returns on earnings

Stage 1:  $EQ\_RET = a + c \text{EAR\_SUR} + \text{RES\_RET}$

Column (2):  $\Delta CDS = \alpha + \beta_1 \text{EAR\_SUR} + \beta_2 \text{EQ\_RET} + \text{controls} + u$

Column (4):  $\Delta CDS = \alpha' + \gamma_1 \text{EAR\_SUR} + \gamma_2 \text{RES\_RET} + \text{controls} + u$

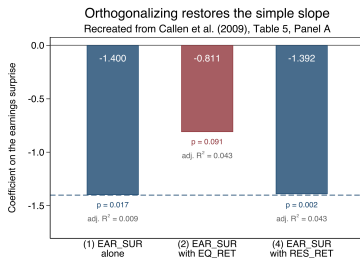
Substituting a pooled-OLS stage 1 into column (2) gives column (4) with  $\gamma_2 = \beta_2$  exactly and  $\gamma_1 = \beta_1 + c\beta_2$ , which equals column (1)'s slope when stage 1 carries column (1)'s controls. The paper reads RES\_RET as equity information other than earnings (p. 1384):

*“Panel B shows that once earnings are defined all-inclusively to include the earnings information about credit risk conveyed by equity prices, earnings provide significant credit risk information for all CDS maturities.”*

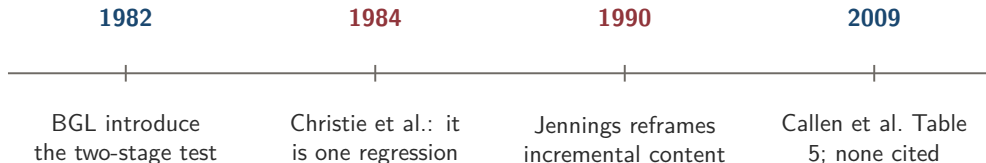
Callen, Livnat, and Segal (2009, p. 1384)

## Table 5 satisfies the identity to rounding

RES\_RET takes  $-0.319$  where EQ\_RET takes  $-0.313$ ; column (4) returns  $-1.392$  where column (1) had  $-1.400$ ; adjusted  $R^2$  is  $0.043$  in columns (2) and (4). Because RES\_RET is a linear combination of EQ\_RET, EAR\_SUR, and a constant, the first equality holds exactly for any linear estimator; the  $0.006$  gap suggests a subgroup first stage or a misprint. The  $-1.400$  comparison is approximate because stage 1 omits column (1)'s controls. Column (2),  $-0.811$  ( $p = 0.091$ ), tests earnings beyond equity returns.



## Table 5 is BGL Panel (A) in one direction



Maroon marks the corrections. Timeline not to scale.

BGL ran both orderings and conceded that Panel (A) “effectively assigns all of that common explanatory power” to the un-orthogonalized variable (p. 29). Callen et al. run one ordering, so the shared variance goes to earnings by construction: “defined all-inclusively” names the assignment, not a finding.

## Footnote 5 ties the sign to random effects

*“One potential explanation for Benkert’s finding is that he uses firm fixed effects in the regression. Since credit ratings are sticky, there is a very high correlation between the firm fixed effects and credit ratings.”*

Callen, Livnat, and Segal (2009, p. 1368, fn. 5)

The authors read the sign reversal as a defect of Benkert’s design; the alternative reading is that the levels relation is identified from between-firm variation, where earnings, size, and the rating co-move.

Footnote 8 (p. 1373) reports that Breusch-Pagan tests “reject firm fixed effects in favor of firm random effects.” That test compares pooled OLS with random effects; random versus fixed is a Hausman (1978) question, not reported. In the within-firm changes design the one-year total is insignificant ( $p = 0.269$ ).

## Firm clustering governs the accrual test

*“Although the accruals coefficient is smaller in absolute value than the cash flow coefficient, suggesting that accrual earnings are less credible to investors than cash flows, the difference is not significant based on clustered standard errors, but is significantly different using conventional robust standard errors ( $p\text{-value} = 0.002$ ).”*

Callen, Livnat, and Segal (2009, p. 1377)

**Units.** Several contracts per firm-quarter (about seven on average) share one ROA, rating, and leverage ratio; 12,242 contracts are at most 2,800 firm-quarters, and the unclustered  $p$ -value treats each as independent.

**Headline.** Page 1364 states “close to 4.5 (2.8) basis points, or by about 9 (5) percent”; the body reports 5 to 6 and 3.5 to 4.5, and 4.5 and 2.8 appear in no table.

## Each question has its own specification

| Question                                | Specification that answers it    | Status in the paper                |
|-----------------------------------------|----------------------------------|------------------------------------|
| Total effect of earnings (H1, headline) | Levels, no RATE                  | Not reported                       |
| Effect incremental to the rating        | Levels with RATE (direct effect) | Table 3, read as the total effect  |
| Earnings beyond returns (window)        | Column (2)                       | Table 5, $p = 0.091$ , set aside   |
| Total window association                | Column (1)                       | Table 5, restated as column (4)    |
| Cash flows versus accruals              | One regression, clustered test   | Reported, then overridden          |
| Maturity interactions                   | Raw or demeaned ROA              | Table 3, correct despite the claim |

Adding a control, centering a regressor, and orthogonalizing a variable each change what a coefficient means; none changes what the data can identify. Two of the three misreadings follow a specification adopted by citation (Aiken and West 1991; the prior CDS studies); the third reproduces an uncited design.

# Summary

- Demeaning a regressor before forming an interaction reparameterizes the model: the interaction coefficient, its standard error, and the fit are unchanged, and the lower variance inflation factor reflects a relabeled column rather than reduced collinearity. The claim is harmless to Table 3, but the reduction it delivers is in a diagnostic, not in the precision of any estimate.
- The S&P rating responds to the earnings the paper studies, so conditioning on it converts the hypothesized total effect into a direct effect, identified only absent unobserved credit information common to rating and premium; footnote 5 suggests the levels result rests on between-firm variation.
- Table 5 reproduces the Beaver, Griffin, and Landsman two-stage residual regression in one direction, without the Christie et al. (1984) correction; the evidence that earnings matter beyond equity returns is the marginal column (2), not the “all-inclusive” column (4).